Thinx

Think, link and analyze human trafficking data

FieldLab 6: Data Liberators
Data Science in Practice 2025

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HUMAN TRAFFICKING RESEARCH

Team member and division of labour

Team tasks:

Thijs: Technical understanding and execution, report writing

Ruben: Team support and guidance, internal and external communication, report writing

Justin: Organisation, note taking, report writing

Bryan: Report leader, report writing

Jeanine: Privacy and security expert, report writing

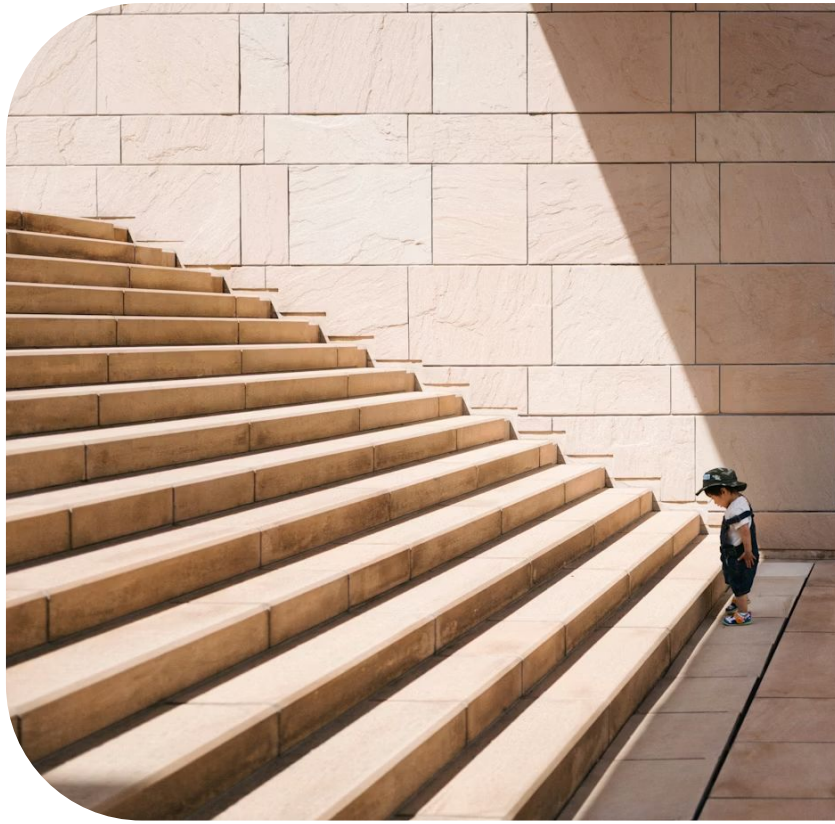
Special thanks to:
Prof. Dr. Mirjam van Reisen
Field Lab Guides: Danial Zemchal & Kai Smits
TA Liam van Dreumel

The challenge

The crisis: Human trafficking in North Africa remains a critical humanitarian issue. Data is often hidden, sensitive, and fragmented.

The data problem: Information exists in "silos", disconnected spreadsheets or local files. Researchers cannot query across datasets without exposing raw, sensitive PII.

The goal: Enable "federated querying", allowing questions like *"Does this trafficker appear in your data?"* without physically or digitally sharing the file.



Project phases

Phase 1: Exploring

Interviewing our stakeholders, creating our technical plan, understanding how FAIR and the EU-GDPR are involved and relevant for our fieldlab,

Phase 2: Tried and tested

Start with creating our pipeline, designing our user interface, create the connections with Allegrograph, whilst always considering the sensitivity of the data and security risk

Final Product: "Thinx"

Shifted to a robust, FAIR-compliant architecture. Focus on data structuring, access control and a UI for metadata management across structured repositories.

Introducing Thinx

A FAIR-Compliant Data Management System

Thinx is a privacy-aware platform developed to bridge the gap between raw data collection and high-level academic analysis.

- **Connect:** Manage multiple disparate datasets (AllegroGraph).
- **Visualize:** Interactive dashboards for file processing and validation
- **FDP integration:** Show FAIR datapoint metadata. Add metadata to repository.

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System architecture

Frontend

Built with **Vue.js**. A responsive SPA that guides researchers through data mapping and visualization without requiring SPARQL knowledge.

Backend API

Powered by **Flask**. Orchestrates the ETL pipeline, handles FAIR metadata fetching, and manages secure connections to the database.

Knowledge Graph

Stored in **AllegroGraph**. A semantic Triple Store allowing for complex relationship mapping (RDF) and federated queries.

Containerized deployment via **Docker**

Secure ETL pipeline

As detailed in our Security Report (Assignment 7), we implemented a strict "Extract-Transform-Load" pipeline to handle sensitive Excel data.

1. **Extraction (Pandas):** In-memory processing of Excel files. Automated validation checks for direct identifiers (names, addresses) immediately upon upload.
2. **Transformation (RDFLib):** Local conversion to RDF Triples based on the Common Data Model (CDM). No external data transfer during this phase.
3. **Loading (Agraph-Python):** Serialized RDF is pushed to the researcher's *private* AllegroGraph repository via encrypted HTTPS/TLS channels.

Least Privilege Principle

"Users should only have the minimum rights required to perform their tasks... reducing risks of data misuse."

Adhering to FAIR-OLR

Thinx extends standard data management by implementing the **FAIR-OLR** framework, ensuring data sovereignty and compliance in federated environments.

FAIR Core

Findable

Access through FDP

Accessible:

Secure authentication via Website.

Interoperable:

RDF & SPARQL standards.

Reusable:

Clear metadata & ontologies.

OLR Extension

Ownership
Data remains with the collector in own DB.

Locale
Data resides in the same location as the program.

Regulatory
Strict adherence to GDPR (Assignments 5 & 6).

User interface strategy

As per Assignment 8, the UI is designed for **non-technical researchers**.

Guided mapping

Visual feedback shows if uploaded data matches the Common Data Model (CDM). Automated warnings for missing fields. Information fields explain functionalities

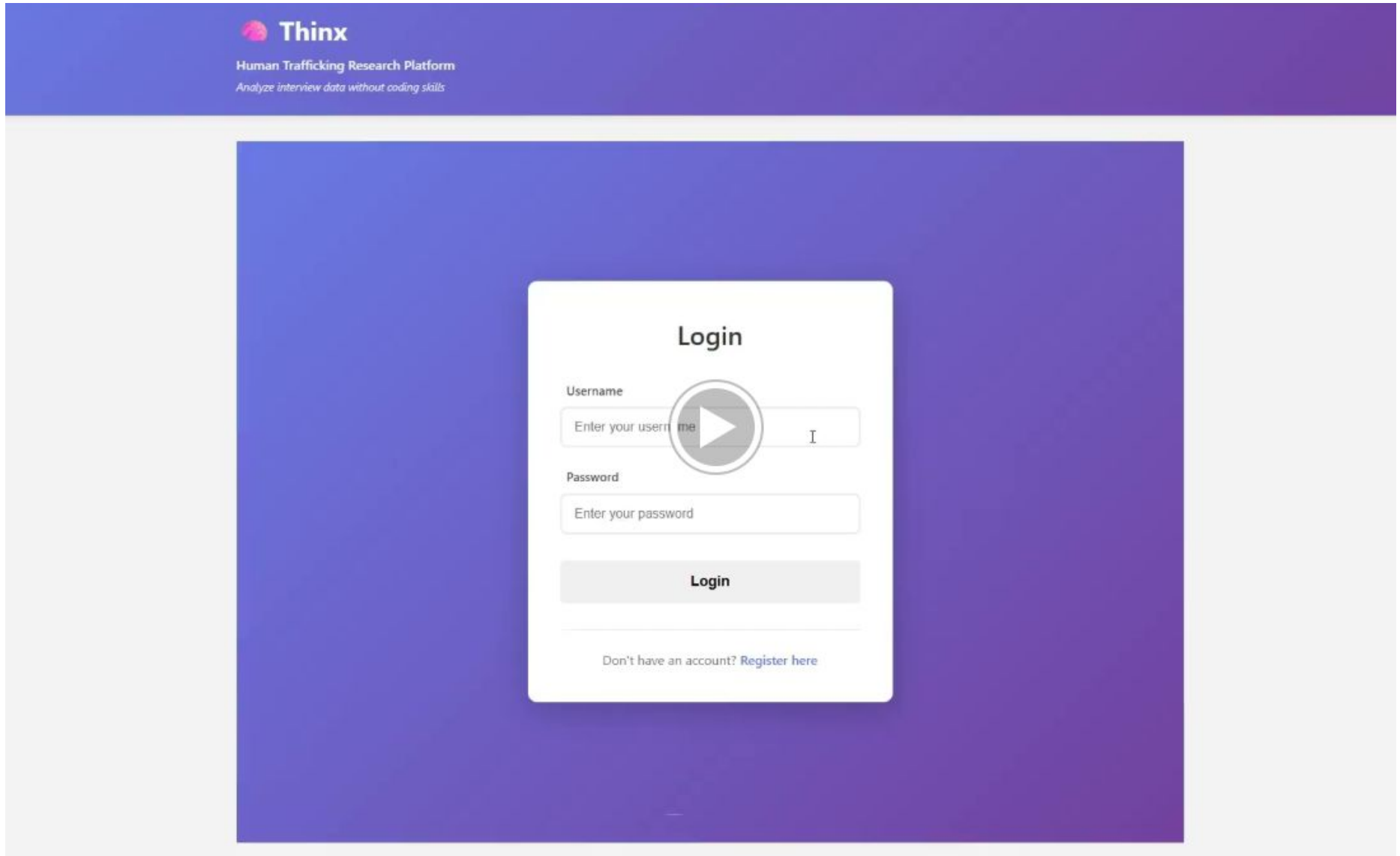
Visual masking

Sensitive fields (PII) are visually masked in the dashboard based on the user's role. However, only the groundwork for the roles has been made at this point.

Connection manager

Simple form to connect to external/internal AllegroGraph repositories. Users input credentials once; the system handles the handshake. Fair data point data is fetched.

Demo of our platform



Thinx

Human Trafficking Research Platform

Analyze interview data without coding skills

Login

Username

Enter your username

Password

Enter your password

Login

Don't have an account? Register here

Documentation Strategy

Thinx is delivered with a comprehensive documentation suite tailored to different stakeholders, ensuring sustainability and easy handover.

For researchers

End-User guide: detailed walkthroughs of the dashboard, mapping tools, and connecting to repositories.

Key Files: README.md, QUICK_START.md

For developers

Technical docs: deep dives into the Flask/Vue stack, Project Structure, and API endpoints.

Key Files: ARCHITECTURE.md, DIAGRAMS.md

For admins

Operations: guides for Docker deployment, environmental configuration, and system migration.

Key Files: DOCKER_USAGE.md, MIGRATION_GUIDE.md

⚡ **Includes QUICK_START.md: Launch the full stack in < 5 minutes.**

Assessment and challenges

Challenges met

- Changing assignment scope
- Managing sensitive data
    - Balancing privacy with research needs
- Non-standard data formats across researchers

How we overcame them

- Multiple stakeholder meetings
- RBAC with isolated repositories
- Local AI-based transformations

What works well

- Strong default privacy
- Clear and controlled access
- Simple column name changes and numerical/categorical transformations using AI

Challenges for the future

- Scaling access control
- Keeping ethical standards consistent
- Implementing more robust AI-based transformations

Thank You

Questions?

Project Team Thinx

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